

π _RL · arXiv 2510.25889 · Oct 5

Archaeologist

11+3 · pack included

Flow VLAs blocked PG → π _RL opens it
→ slow-sim routes around via RFT

Kanta not seated — all five packs built

Spine: unlock PPO on flow VLAs; BEHAVIOR still prefers RFT

Chen / Liu / Zhang et al. · RLinf · ICML-style (TENTATIVE)

Club spine · Oct 5

Unlock PPO on flow VLAs · BEHAVIOR still prefers RFT

BLOCKED

Flow VLAs

intractable $\log \pi$
+ deterministic ODE
→ no PG / no explore

π_{RL} OPENS

Flow-Noise / SDE

tractable Gaussian
+ PPO/GAE on $\sim 300M$
expert (VLM frozen)

FAST SIM

LIBERO / ManiSkill

+27-31 pp ID avg
few-shot+RL > full SFT
partial env OOD

SLOW SIM

BEHAVIOR / OmniG

Comet cites π_{RL}
then uses RFT
sample-efficiency gate

When to use π_{RL}

- Parallel sparse-success sim available
- OpenPI π_0 / $\pi_{0.5}$ (or GR00T) already SFT'd
- Want ID specialists beyond demos
- Can afford 8×H100-class + 64-320 envs
- Real deploy = sim RL → ODE eval (not real RL)

When to stay on RFT

- Slow household sims (BEHAVIOR)
- Heterogeneous GPU / RT-core split pain
- Need coverage flywheel, not on-policy
- Comet path: reject → SFT (Q 0.224 → 0.345)
- 2026: compose RFT then short π_{RL} burst

Lineage arc · Archaeologist

flow VLAs blocked PG → π_{RL} opens it → slow-sim routes around via RFT



Priors absorbed (CLEAR)

π_0 / $\pi_{0.5}$ · Lipman FM / Rectified Flow · ReinFlow · Flow-GRPO · DPPO

PPO / GAE · RLinf / RLinf-VLA · Song PF-ODE↔SDE · 3DGS (Real2Sim)

Contemporaries: SimpleVLA-RL, RL4VLA, FPO (not compared) · Subsequent: Comet RFT, BEHAVIOR'25 forks

Do not fabricate: no BEHAVIOR win · no ICML acceptance · no real online RL at scale

Flow-Noise vs Flow-SDE · two ways to unlock PPO

Both train stochastic · both deploy as deterministic ODE (noise discarded / SDE off)

Flow-Noise

ReinFlow-inspired · one-layer MDP

- ▶ Learnable $\sigma_{\theta'}(A^{\tau}, o) \rightarrow$ Gaussian denoise step
- ▶ Exact joint log-prob of denoise chain (eq. 5)
- ▶ `joint_logprob=True` · entropy ~ 0.005
- ▶ Noise net discarded after fine-tune
- ▶ Often wins MetaWorld (π_0 85.8 vs 78.1)
- ▶ Cost: full denoise recompute for likelihood

Flow-SDE

Flow-GRPO / DPPO · two-layer + hybrid

- ▶ ODE $\rightarrow$ equivalent SDE (Song PF-ODE)
- ▶ Noise $\propto \sqrt{\tau/(1-\tau)}$; Gaussian step likelihood
- ▶ Two-layer MDP: denoise inner + env outer
- ▶ Hybrid: 1 SDE step / env · rest ODE $\rightarrow \sim 2\times$
- ▶ Default efficiency pick (paper preference)
- ▶ Often wins CALVIN $\pi_{0.5}$ Len-5 (87.0)

Shared recipe

few-shot SFT $\rightarrow$ freeze VLM $\rightarrow$ PPO/GAE on $\sim 300M$ action expert · chunk reward · RLinf · $8 \times H100$

Control note: chunk length H = credit-assignment knob · not classical planner

Control note · not classical planner

Flow → MDP

Stochastic via Flow-SDE
or Flow-Noise train path
Deploy = ODE Euler

Hybrid sampler

1 SDE step / env step
rest ODE → $\sim 2\times$ faster
vs full two-layer

PPO on expert

$\sim 300\text{M}$ action expert
VLM frozen default
chunk-level reward

Chunk length H = credit-assignment knob

Paper: π_0 LIBERO $H=50$ · $\pi_{0.5}$ $H=10$ · ManiSkill $H=8$ · CALVIN/MW $H=5$

Larger chunks $\uparrow$ SFT baseline but $\downarrow$ explained variance / RL ceiling

Empiricist: prefer Comet-like mid horizons + receding H' over cargo-cult $H=50$

Online RL (π _RL) vs RFT (Comet) · product fork

Axis Needs	Online RL (π _RL) log π + exploration	RFT (Comet) success filter only
Sample efficiency	low	higher for slow sims
Compute topology	tight env $\leftrightarrow$ train loop	embar. parallel eval OK
Ceiling	can exceed demos	capped by reachable modes
Cross-task OOD	weak (ML45)	also limited
BEHAVIOR'25	cited, not used	Q 0.224 $\rightarrow$ 0.345
2026 upside	if faster eval arrives	safe default flywheel

Table 1 · ID average success % · tattoo these

Four suites averaged · Δ vs few-shot SFT

Model	Method	LIBERO	ManiSkill	MetaWorld	CALVIN	Avg	Δ
π_0	SFT	57.6	38.4	50.8	57.5	51.1	—
π_0	Flow-SDE	96.1	78.8	78.1	61.7	78.7	+27.6
π_0	Flow-Noise	97.6	77.8	85.8	59.9	80.3	+29.2
$\pi_{0.5}$	SFT	77.1	40.1	43.8	61.3	55.6	—
$\pi_{0.5}$	Flow-SDE	97.9	90.9	70.7	87.0	86.6	+31.0
$\pi_{0.5}$	Flow-Noise	98.3	89.7	66.1	84.5	84.7	+29.1

π_0 SFT→Noise

51.1 → 80.3 (+29.2)

$\pi_{0.5}$ SFT→SDE

55.6 → 86.6 (+31.0)

Best avg

$\pi_{0.5}$ Flow-SDE 86.6

OOD honesty · what transfers

TRANSFERS

- ▶ ManiSkill visual / layout / exec shifts
- ▶ CALVIN ABC→D scene transfer
- ▶ Partial relative OOD gains ($\sim +100\%$)
- ▶ Preserves SFT OOD skills (less forgetting)

DOES NOT

- ▶ MetaWorld ML45 novel tasks — oscillates
- ▶ Cross-task skill invention via RL
- ▶ Open-world discovery narrative
- ▶ Visual OOD ceiling (frozen VLM)

Reading for Reviewer + Empiricist

RL = action-level refinement on seen skills

Not a substitute for multi-task demos or stage structure

BEHAVIOR implication: recovery / temporal efficiency — not 50-task invention

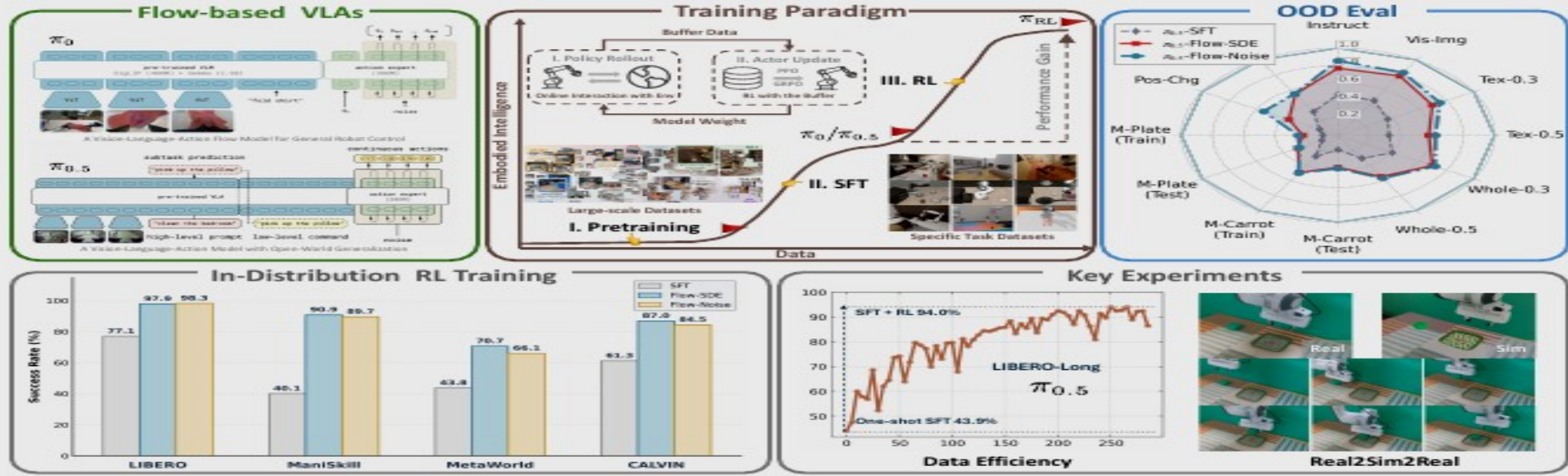


Figure 1. π_{RL} : An online RL framework for flow-based VLAs. Incorporating two solutions, Flow-Noise and Flow-SDE, π_{RL} enhances the performance and generalization of SFT-aligned models across extensive ID benchmarks and OOD settings. Refined with RL, few-shot SFT policies achieve performance comparable to full dataset baselines. Additionally, we facilitate seamless zero-shot sim-to-real transfer by constructing a simulator with 3D Gaussian Splatting as the rendering engine to narrow the visual domain gap.

Abstract

Vision-Language-Action (VLA) models enable robots to understand and perform complex tasks from multimodal input. Although recent work explores using reinforcement learning (RL) to automate the laborious data collection process in scaling supervised fine-tuning (SFT), applying RL to large-scale flow-based VLAs (*e.g.*, π_0 ,

$\pi_{0.5}$) remains challenging due to intractable action log-likelihoods raised from flow matching. We address this challenge with π_{RL} , featuring two technical approaches: (1) **Flow-Noise** models the denoising process as a discrete-time MDP with a learnable noise network for exact log-likelihood computation. (2) **Flow-SDE** integrates denoising with agent-environment interaction, formulating a two-layer MDP that employs ODE-to-SDE conversion for efficient RL exploration. We evaluate π_{RL} across various benchmarks, with experiments demonstrating that RL yields significant performance improvements in both in-distribution and out-of-distribution settings.

¹Tsinghua University ²Peking University ³Institute of Automation, Chinese Academy of Sciences ⁴Carnegie Mellon University ⁵Infinigence AI ⁶Zhongguancun Academy [†]Work done at Tsinghua University * Equal Contributions. Correspondence to: Yu Wang <yu-wang@tsinghua.edu.cn>, Chao Yu <zoeyuchao@gmail.com>.

(DPO) (Rafailov et al., 2023) with stage-based sparse rewards. RLinf-VLA (Yu et al., 2025; Zang et al., 2025) provides a unified and efficient framework for scalable RL training of VLA models. These works demonstrate the effectiveness of RL fine-tuning VLA models.

2.3. RL Fine-tuning for Flow Models

Integrating RL with flow models is a promising way to transcend the limitations of imitation learning. To this end, Flow-GRPO (Liu et al., 2025b) converts the deterministic ODE into an equivalent SDE to enable stochasticity exploration, a foundation upon which subsequent works like Mix-GRPO (Li et al., 2025b) and TempFlow-GRPO (He et al., 2025) further accelerate training through hybrid ODE-SDE rollouts. ReinFlow (Zhang et al., 2025) injects learnable noise into the flow path and transforms it into a discrete-time Markov process with a tractable likelihood for stable policy gradient updates. Flow policy optimization (FPO) (McAllister et al., 2025) reframes policy optimization as maximizing the advantage-weighted ratio of the conditional flow matching loss.

3. Preliminary

3.1. Problem Formulation

We formulate the task as an MDP, defined by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P_0, P_{\text{ENV}}, R_{\text{ENV}}, \gamma)$. The state $s_t \in \mathcal{S}$ is defined as the robot observation $\mathbf{o}_t$ and P_0 denotes the initial state distribution. Given the state, the flow policy predicts an action $a_t \sim \pi(\cdot|s_t) \in \mathcal{A}$, resulting in the state transition $s_{t+1} \sim P_{\text{ENV}}(\cdot|s_t, a_t)$ and a reward $R_{\text{ENV}}(s_t, a_t)$. The objective is to learn a policy π_θ that maximizes the expected γ -discounted return over a horizon of $T + 1$:

$$\mathcal{J}(\pi_\theta) = \mathbb{E}_{\pi_\theta, P_0} \left[\sum_{t=0}^T \gamma^t R_{\text{ENV}}(s_t, a_t) \right]. \quad (1)$$

With the policy gradient surrogate (Williams, 1992), the gradient of the return expectation can be approximated from sampled trajectories:

$$\nabla_\theta \mathcal{J}(\pi_\theta) = \mathbb{E}_{\pi_\theta, P_0} \left[\sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t|s_t) A(s_t, a_t) \right]. \quad (2)$$

The advantage function, $A(s_t, a_t) = Q(s_t, a_t) - V(s_t)$, measures the relative merit of the action value $Q(s_t, a_t)$ over the state value $V(s_t)$, providing a low-variance signal for the policy update.

3.2. Flow-based Vision-Language-Action Model

A flow-based VLA model π_θ is designed to map the observation $\mathbf{o}_t$ comprising RGB images, language tokens, and

robot proprioception to a sequence of H future actions $\mathbf{A}_t = [a_{t,0}, \dots, a_{t,H-1}]$, formulated as $p(\mathbf{A}_t|\mathbf{o}_t)$. Within the model, the VLM extracts features from the visual and language inputs, while the flow matching expert is tasked with generating the actions. Specifically, the model learns a conditional vector field $\mathbf{v}_\theta$ that transforms a standard Gaussian noise distribution into the target action $\mathbf{A}_t$. This is achieved by minimizing the Conditional Flow Matching (CFM) loss, which aligns the predicted vector field $\mathbf{v}_\theta$ with the ground-truth vector field $\mathbf{u}$:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{\tau, p(\mathbf{A}_t, \mathbf{o}_t), q(\mathbf{A}_t^\tau|\mathbf{A}_t)} \left[\|\mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t) - \mathbf{u}(\mathbf{A}_t^\tau|\mathbf{A}_t)\|_2^2 \right]. \quad (3)$$

Here, the conditional probability path $q(\mathbf{A}_t^\tau|\mathbf{A}_t)$ generates a noisy action¹ $\mathbf{A}_t^\tau = \tau \mathbf{A}_t + (1 - \tau)\epsilon$ from an action $\mathbf{A}_t$, random noise $\epsilon \sim \mathcal{N}(0, I)$, and a continuous time $\tau \in [0, 1]$ in flow matching. For this specific path, the corresponding ground-truth vector field is defined as $\mathbf{u}(\mathbf{A}_t^\tau|\mathbf{A}_t) = \mathbf{A}_t - \epsilon$.

During the inference, the action sequence is generated by first sampling a noise vector $\mathbf{A}_t^0 \sim \mathcal{N}(0, I)$, which is further iteratively refined by integrating the learned vector field $\mathbf{v}_\theta$ over a fixed number of steps based on the forward Euler method: $\mathbf{A}_t^{\tau+\delta} = \mathbf{A}_t^\tau + \mathbf{v}_\theta(\mathbf{A}_t^\tau, \mathbf{o}_t) \cdot \delta$.

4. Methodology

Existing VLA-RL approaches leverage base models such as OpenVLA for discrete actions and OpenVLA-OFT for continuous actions. To compute the action log-likelihood $\log \pi_\theta(a_t|s_t)$, discrete models (Liu et al., 2025a) apply softmax to the output logits, while continuous models (Li et al., 2025a) treat the action as a Gaussian distribution, employing a prediction head to estimate the variance. As for the flow-based VLAs, directly computing the exact likelihood (Hutchinson, 1989) is inaccurate with few denoising steps. Moreover, the deterministic nature of its ODE sampling process precludes exploration, making its implementation within RL non-trivial. To this end, we propose Flow-Noise and Flow-SDE, two technical approaches that make flow-based VLAs amenable to RL.

4.1. Flow-Noise

Inspired by Reinflow (Zhang et al., 2025), we incorporate a learnable noise network into the flow matching denoising process and solve the problem within the standard one-layer MDP framework detailed in Sec. 3.1. By modeling the denoising stage as a discrete MDP, we can directly compute the log-likelihood of the denoised sequence, enabling equivalent policy optimization via RL.

¹ $\mathbf{A}_t^\tau$ incorporates two temporal indices, t denotes the discrete time step for environment interaction and τ represents the continuous time variable in flow matching.

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Reviewer verdict

Strong same-task mastery

+ partial env-shift robustness

Don't overread as open-world discovery

ACCEPT-LEARNING

methods+systems
if sample-efficiency
+ multi-seed improve

WHAT IT SHOWS

ID polish after
few-shot SFT;
env/visual OOD partial

WHAT IT DOESN'T

novel-task invention;
real online RL;
iso-compute vs SFT

Archaeologist close

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π_{RL} : Online RL Fine-tuning for Flow-based Vision-Language-Action Models

Kang Chen^{2,6,*} Zhihao Liu^{3,6,*} Tonghe Zhang^{4,*} Zhen Guo⁵ Si Xu⁵ Hao Lin⁵
 Hongzhi Zang¹ Xiang Li⁵ Bingwen Wei¹ Jiakai Zhou¹ Quanlu Zhang⁵
 Zhaofei Yu² Guoliang Fan³ Tiejun Huang² Yu Wang^{1†} Chao Yu^{1†}

<https://github.com/RLinf/RLinf> <https://huggingface.co/RLinf>



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